Mohamed Elmugtaba Mohamed

BIOINFORMATICS PRACTITIONER | GENOMIC DATA ANALYSIS & REPRODUCIBLE PIPELINES | PHARMACIST

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PROFESSIONAL SUMMARY

> *A pharmacist who retrained in computational biology and builds reproducible, tested genomic workflows.*

Pharmacist with 7+ years of professional experience, 6+ of them across pharmacy practice, GMP pharmaceutical manufacturing and medical and scientific sales. Builds Nextflow DSL2 pipelines on Linux for germline variant calling and annotation (GATK4, ClinVar, SnpEff, trio filtering) and RNA-seq, with Git, modular code, nf-test automated testing, GitHub Actions CI and Docker containers, and has hands-on single-cell RNA-seq analysis in Python (Scanpy). GMP production work (batch-record approval, 100% compliance on every product release) built the documentation and quality habits that reproducible analysis depends on; scientific sales built clear communication of evidence to physicians. Seeking an entry-level bioinformatics role in genomics or precision medicine.

BIOINFORMATICS PROJECTS

1. Germline Variant Calling & Annotation Pipeline (BAM → VCF)

↗ GitHub

`Nextflow DSL2` `GATK4` `bcftools` `SnpEff` `ClinVar` `nf-test` `GitHub Actions` `Docker`

> **Output:** Joint-genotyped trio VCF of germline SNPs and indels, annotated with ClinVar and SnpEff, plus a table of protein-altering and de novo candidates; QC in a MultiQC report; 18 nf-tests passing in CI.

SCALE	Family trio (3 samples) · chromosome 20 slice · hg19/b37 · starts from aligned BAMs.
METHODS	HaplotypeCaller (GVCF) → GenomicsDBImport → GenotypeGVCFs → coordinate shift → ClinVar (bcftools annotate) → SnpEff → protein-altering and de novo filters; samtools and bcftools stats → MultiQC. nf-schema validation, fail-fast input checks, retry on out-of-memory exits.
TESTS	Positive and negative controls for the coordinate-shift, ClinVar and filter steps.
SOURCE	Dataset and calling steps adapted from Seqera's Nextflow for Science: Genomics course (credited in the repo); annotation and trio filtering added independently.

2. Reproducible Human RNA-seq Pipeline (FASTQ → counts)

↗ GitHub

`Nextflow DSL2` `FastQC` `fastp` `STAR` `featureCounts` `MultiQC` `Docker`

> **Output:** Raw FASTQ → sorted BAM → gene-level count matrix, with all QC aggregated in one MultiQC report.

SCALE	Human chromosome 22 (GRCh38, Ensembl 110) · single- or paired-end reads.
METHODS	STAR index → FastQC → fastp → STAR → featureCounts → MultiQC; Docker containers on a local executor (containerised, not yet run on HPC or cloud); fail-fast checks on the samplesheet and references.

3. COVID-19 vs Control PBMC Single-Cell RNA-seq Analysis

↗ GitHub

`Python` `Scanpy` `AnnData` `Scrublet` `Scanorama` `gseapy`

> **Output:** QC-filtered 9,000 cells to 5,589; compared three batch-integration methods; clustered (Louvain, 9 clusters), annotated by reference label transfer and Enrichr markers, and compared COVID-19 with control cells within clusters.

SCALE	6 samples (3 COVID-19, 3 controls) · 10x Genomics · 1,500 cells each.
METHODS	Mito/ribo QC, sex and cell-cycle checks, doublet removal; MNN, ComBat, Scanorama; Wilcoxon, t-test and logistic-regression DE; Enrichr and GSEA.
SOURCE	NBIS project-based scRNA-seq workshop.

4. Single-Cell & Spatial Analyses `TUTORIAL-BASED`

Visium + scRNA-seq integration: human lymph node QC, clustering and spatially variable genes (SpatialDE); two mouse-brain sections integrated with Scanorama and labelled from a cortex reference. ↗ GitHub

Scarf: 10x 5K PBMC clustering, and projection of 14K interferon-stimulated PBMCs onto a 15K control reference with label transfer. ↗ GitHub

TECHNICAL SKILLS

LANGUAGES & ENVIRONMENT

Python · R (ggplot2) · Bash · Linux / Unix · Jupyter

WORKFLOWS & PRACTICES

Nextflow DSL2 · nf-test · nf-schema · nf-core* · Docker · Conda · Git / GitHub · GitHub Actions CI

NGS, VARIANT ANALYSIS & QC

GATK4 · HaplotypeCaller · GenomicsDBImport · GenotypeGVCFs · bcftools · SnpEff · ClinVar · Samtools · STAR · featureCounts · FastQC · fastp · MultiQC

SINGLE-CELL & SPATIAL

Scanpy · AnnData · Scanorama · ComBat / MNN · Scrublet · gseapy · Scarf* · SpatialDE*

* course- or tutorial-level

EDUCATION

BSc Pharmacy

National Ribat University

Khartoum, Sudan · 2010 – 2015

Coursework: biochemistry, microbiology, pharmacology, clinical pharmacology, analytical chemistry.

CERTIFICATIONS

PIPELINES & NGS

- **Nextflow Training Week 2026-Q1: 5 certificates**
  Hello Nextflow · Nextflow Run · Hello nf-core · Nextflow for RNAseq · Nextflow for Genomics
  Seqera · Feb 2026

BIOINFORMATICS

- **Bioinformatics for Biologists: An Introduction to Linux, Bash Scripting, and R**
  Wellcome Connecting Science / FutureLearn · Jul 2023 · 100%
- **scRNA-seq workshop (project-based)**
  NBIS, National Bioinformatics Infrastructure Sweden
- **Python**
  Kaggle · Feb 2022

TRAINING & TRANSITION

Self-directed retraining in computational biology

2022 – 2024

Career break after a family relocation from Sudan during regional instability, used for structured self-study.

- ✓ **Foundations:** Python, Kaggle (Feb 2022)
- ✓ **Portfolio:** single-cell and spatial analyses on GitHub (Jul–Sep 2022), including the NBIS COVID-19 workshop project
- ✓ **Visualisation:** R and ggplot2 scripts (Jul 2023)
- ✓ **Coursework:** Linux, Bash scripting and R for biologists, Wellcome Connecting Science, 100% (Jul 2023)

Pipeline engineering

2025 – Present

Relocated to Riyadh (Oct 2025); structured training and pipeline building alongside the job search.

- ✓ **Seqera (Feb 2026):** 5 Nextflow certificates, including RNA-seq, genomics and nf-core
- ✓ **Built:** RNA-seq pipeline (Feb 2026) and GATK4 germline variant-calling pipeline with nf-test and CI (Sep–Oct 2026), extended with ClinVar/SnpEff annotation and trio filtering (Oct 2026)
- ✓ **Cloud:** Google Cloud Digital Leader badge, Google Skills (Jan 2026)
- ✓ **In progress:** Integrating Generative AI into Data Workflows (Google Skills)

PROFESSIONAL EXPERIENCE

Key Account & Operations Coordinator

2024 – 2025

Dukhan Group · Qatar

- Built and maintained operational trackers (contract status, SLA compliance, recovery outcomes), checking the data across business units.
- Found the root causes of workflow bottlenecks between departments; primary liaison between clients and internal teams to resolve service failures.

Scientific Sales Officer

2021 – 2022

Salmawit Co. Ltd · Sudan

- Turned published clinical study data into evidence-based communication for physicians on efficacy, safety and appropriate prescribing across a GIT/IBS portfolio.
- Planned and delivered CME events and clinical seminars in gastroenterology at least once a quarter; ran a structured call cycle of 120 physicians.

Production Pharmacist

2019 – 2020

Blue Nile Pharmaceutical Factory · Sudan

- Reviewed and approved batch production and manufacturing records (BPR/BMR) before release, with 100% GMP compliance on every product release.
- Planned production schedules against sales forecasts and coordinated with procurement on raw materials.
- Supervised end-to-end production and a 90-person team, coordinating QA, QC and maintenance as monthly output roughly quadrupled (15–17M → 60–65M tablets/month).

Medical Representative (Territory Lead)

2017 – 2018

Aurobindo Pharma · Sudan

- Led the launch of a new antibiotic brand in a high-priority territory and secured a government formulary listing under the national government insurance scheme; gave scientific presentations and roundtables for key opinion leaders.

Medical Representative

2016 – 2017

Bioderma · Sudan

Pharmacist

2015 – 2016

Wenji Pharmacy & Sudan Military Hospital · Sudan

- Dispensing, dose verification and drug-interaction screening; patient counselling on adherence and side effects.

LANGUAGES Arabic (native) · English (full professional proficiency) · **STATUS** Valid, transferable Iqama (KSA) · **AVAILABILITY** Immediate

Updated October 2026